

ADAM Optimisation of Hotwire Cutter Trajectory

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Abstract – Hot-Wire Cutting can rapidly produce prototype components for small unmanned aircraft but manufacturing accuracy is hard to control due to the complex interactions at the cutting interface. In this report methods are presented that aims to understand and correct this problem?

I. INTRODUCTION

Hot-wire cutting (HWC) is a subtractive manufacturing method that produces components that can be described by a ruled surface. Hot-wires remove material from the bulk by ablation or melting in a region close to the wire. The size of the region subject to ablation or melting is dependent of a large number of independent parameters. Finding the size of the removed region is a multivariant problem that is commonly taken to be fixed for large manufacturing operations where the resulting deviation from the desired shape is insignificant. When used in aerospace applications the exact geometry becomes more important, this is particularly critical for small scale aircraft where the relative scale of the cutting errors become unacceptably large and can have dramatic effects on flight characteristics [1]. Central to this challenge is the manufacture of thin geometries in these cases controlling the amount of thermal energy left in the material after the cut is completed becomes important, for the HWC to complete the cut two passes must be made near the thin geometry, the heat left in the material from the previous cut increases the amount of ablation on the second pass dramatically altering the cutting kerf.

This paper aims to realize a method for optimizing the wire trajectory such that the manufactured shape more closely matches the design intent. As a corollary this paper also aims to demonstrate a new method for predicting the material behaviour under cutting conditions.

In this paper a 4-axis CNC hotwire is used (CNC Multitools 1610S). This machine allows for independent control of both the position and velocity of each of the Four axes, and importantly the wire heating is manually set [2].

II. MOTIVATION

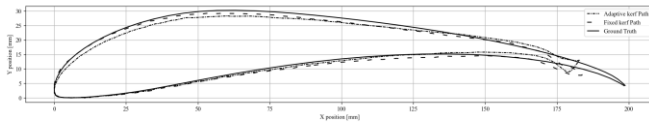


Figure 1. Processed surface geometry from foils manufactured using existing methods, S1223 Foil [3], designed chord length of 200 mm, also shown is an existing optimisation method.

Figure 1 shows processed data from a set of Two foils manufactured using existing methods to generate the machine tool paths. The scans were performed with the EINSKAN HX 3D scanner with 0.2 mm resolution. The resulting data shows how the manufactured geometry deviates from the desired geometry (solid line).

As previously stated small deviations in foil geometry can have disproportionate effects on flight performance and characteristics. A central goal to meet design specification results in tight margins and the need for the manufactured geometry to closely match the theoretical performance, we can ensure this by more closely matching the manufactured geometry to the design geometry.

Figure 2 shows that as a result of manufacturing inaccuracies the manufactured foil geometry has a 10% reduction of lift generated at 0° Angle of Attack and $Re = 130,000$.

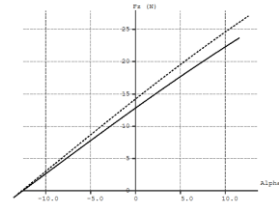


Figure 2. Comparison of theoretical performance of designed foil geometry (dashed) and manufactured foil geometry (solid). $V_0 = 10 \text{ ms}^{-1}$ at $Re = 130,000$ [5].

III. AIMS & OBJECTIVES

The central aim of this paper is as follows:

Develop a method to optimise hotwire trajectory to minimise deviations from the designed surface shape.

This aim has been broken into a set of quantifiable objectives used to track progress:

OB1. Undertake a testing campaign to quantify magnitude and types of errors for standard tool paths, and for the existing method for correcting these errors

OB2. Research, develop and implement the a new approach.

OB3. Compare manufacturing quality, computational requirements, and perform trade off analysis.

OB4. Perform wind tunnel testing to compare aerodynamic performance.

IV. METHODOLOGY

The method developed is into 3 parts:

A. Material modelling.

For this method to work the material must first robustly predict the resulting shape from a given wire trajectory. To simplify this problem it is assumed that at the instant ablation at a point ceases the surface specific energy will be a known material-dependent value, If a given surface is exposed to more thermal energy than would be required to bring it to this critical surface specific energy the surface will be over ablated and vice versa, it is now only necessary to calculate the cumulative surface thermal exposure at each location to determine if the final shape will be accurately produced.

B. Thermal modelling

To compute the surface exposure both the wire trajectory and surface geometry are discretised, each panel is defined by two points and a vector.

To simplify computation a binary mask is used to track which panels have valid interaction, the mask is formed by taking the sign of the vector cross product of every combination of wire panel and surface panel. *Figure 3* shows the results of this operation, black regions show what wire positions will result in ablation of which surface positions.

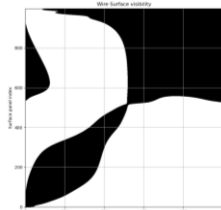


Figure 3. Wire, Surface visibility, Black regions describe regions on the surface that are visible at a chosen wire position.

The sign of value in the resulting matrix describes if the wire panel originates from a point to the 'left' or 'right' of the surface panel vector. The sign chosen is dependent on a combination of foil orientation and surface cutting direction, in our case foils are cut in an anticlockwise direction from foil TE -> LE -> TE (Trailing edge, Leading edge).

Each point combination is now weighted according to the relative angle between panel vectors and the separations between panels and size of each panel.



Figure 4. Wire, Surface coupling, Brighter colours represent regions of stronger coupling.

Figure 4 shows the exposure of each surface panel as a result of a wire panel, it can be seen that most regions experience similar heating but some regions near the trailing edge experience significantly more heating as a result of the foils geometry.

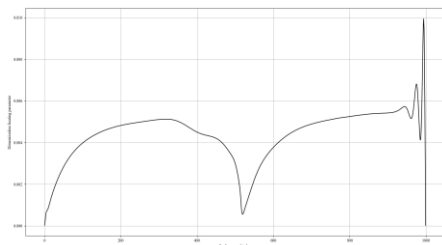


Figure 5. Panel total exposure.

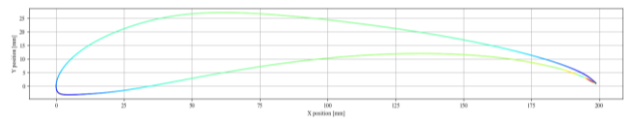


Figure 6. Superimposed panel total exposure.

Taking the sum of exposures for each surface panel we arrive at *Figure 5*, it is clear that although many regions experience similar heating some regions are particularly under or over exposed. Superimposing this data onto the foil geometry we arrive at *Figure 6*. Comparing *Figure 6* to *Figure 1* it can be seen that regions with higher exposure are also the regions that most deviate from the desired geometry.

C. ADAMs path optimisation

Using the developed thermal modelling method we can now begin optimising the wire trajectory, the goal of the optimisation is for all panels to receive equal exposure, the means that the total ablation will not be well known and a predictable kerf width can be identified.

ADAMs has been chosen as it has been shown to efficiently converge for noisy objective functions in a large parameter space such as the one[6].

V. EXPECTED RESULTS

It is expected that by using ADAMs to optimise the wire trajectory the desired surface geometry can be preserved and the expected performance can be realised.

VI. REFERENCES

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